sophisticated technology to make atom bombs, but India demonstrated that we could make them indigenously. I saw this as a similar opportunity for a new generation of entrepreneurs to demonstrate that India could be a world leader in the field of vaccinations."

Adar Poonawalla viewed the challenge the pandemic posed somewhat differently. As he put it, "We had two choices, either to sit and wait as a manufacturer for a technology to prove itself or enter into collaborations with multiple innovators and academia such as Oxford, Novavax and others to develop the vaccine apart from building facilities to stockpile production. The latter option came with a lot of risk and challenges. We had to raise capital of around $1 billion to expand our facilities, then get permissions to make the vaccines and after that supply them in large numbers. But we decided that speed was essential and went ahead with it."

Like Ella, Adar Poonawalla too had done what he calls "future planning" to gear up his firm for a pandemic-like eventuality. Since he took over as CEO in 2011, he had expanded Serum's facility to build a variety of new vaccines, including installing specialised equipment for their speedy manufacture in large quantities. He calculated that he would be able to quickly rejig these to make any vaccine that was needed. His company already had ties with AstraZeneca, the British-Swedish pharma giant, to make a vaccine for malaria. So, when AstraZeneca tied up with Oxford University to develop a Covid-19 vaccine, Poonawalla says, "It was obvious that they should be our first partner." Thus, the world was introduced to Covishield, as he called the Indian version of the vaccine. Subsequently, Adar would also tie up with Novavax and other companies to make Covid vaccines apart from asking his own team to work on an indigenously developed one.

But both Ella and Poonawalla initially decided to invest in vaccines that used conventional and established technologies rather than the newer gene-based ones. Ever since the smallpox vaccine was discovered by Edward Jenner in the 18th century, scientists have used his technique of injecting inactivated versions of the disease-carrying virus to develop vaccines that provide immunity against it. Simply put, if a person is injected with or swallows a tiny amount of what is known as a 'killed' or inactivated virus (in which its replicating ability is chemi-

POONAWALA TOOK THE HUGE FINANCIAL RISK OF STOCKPILING MILLIONS OF VIALS OF COVISHIELD EVEN BEFORE THE RESULTS OF ITS HUMAN CLINICAL TRIALS WERE IN, A GUTSY DECISION THAT PAID OFF

THE CENTRE STEPPED IN TO FAST-TRACK THE VACCINE CLEARANCE PROCESS WITHOUT COMPROMISING ON REGULATORY RIGOUR AND INTEGRITY

